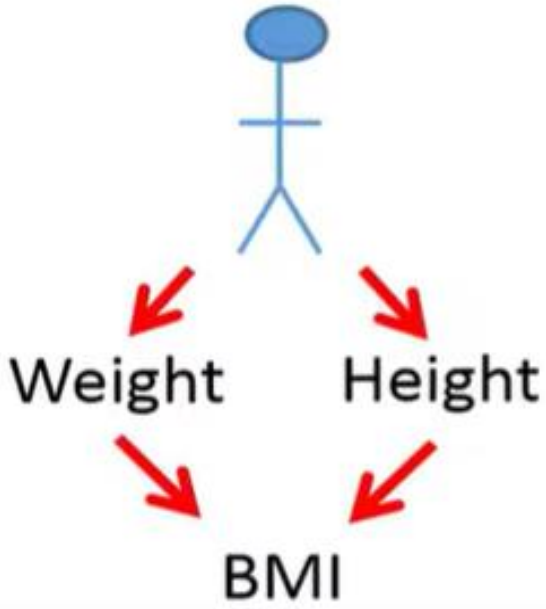
Principal Component Analysis (PCA)

PCA

- Combining variables
- PCA – basics
- Its applications

Combining variables

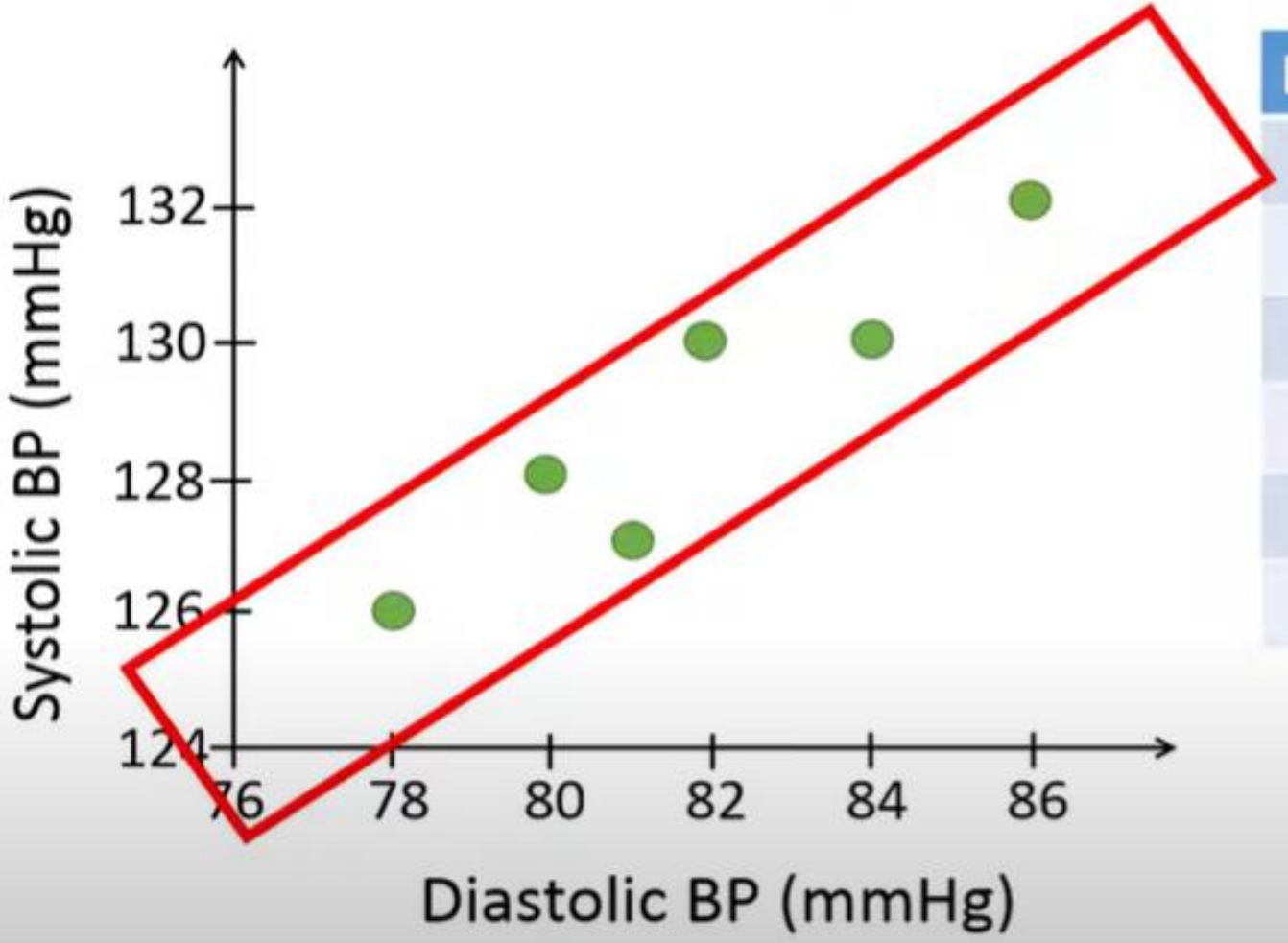


Cholesterol = Weight + Height

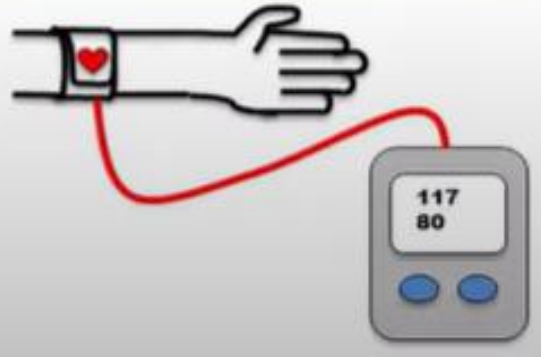
Cholesterol = BMI

$$BMI = \frac{Weight_{kg}}{Height_{m}^{2}}$$

Combining variables



Diastolic BP	Systolic BP
78	126
80	128
81	127
82	130
84	130
86	132



Combining variables

Diastolic BP	Systolic BP
78	126
80	128
81	127
82	130
84	130
86	132

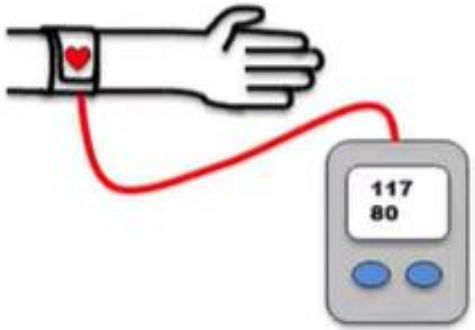
$$Y = \alpha_1 X_1 + \alpha_2 X_2$$

$$BP = \boxed{\alpha_1} DBP + \boxed{\alpha_2} SBP$$

$$BP = \boxed{0.8} DBP + \boxed{0.6} SBP$$

Combining variables

Diastolic BP	Systolic BP	Mean BP
78	126	
80	128	
81	127	
82	130	
84	130	
86	132	



$$BP = \alpha_1 DBP + \alpha_2 SBP$$

$$\boxed{BP_1 = \frac{DBP_1 + SBP_1}{2}}$$

$$BP_1 = \boxed{\frac{1}{2}} DBP_1 + \boxed{\frac{1}{2}} SBP_1$$

$$BP_1 = 0.5 DBP_1 + 0.5 SBP_1$$

Combining variables

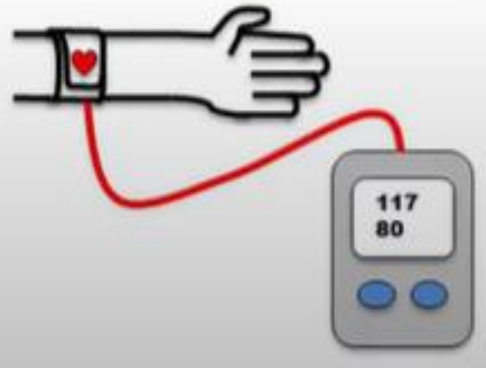
Diastolic BP	Systolic BP	Mean BP
78	126	102
80	128	104
81	127	104
82	130	106
84	130	107
86	132	109

$$BP = \alpha_1 DBP + \alpha_2 SBP$$

Combining variables

Diastolic BP	Systolic BP	Mean BP	Sum BP
78	126	102	204
80	128	104	208
81	127	104	208
82	130	106	212
84	130	107	214
86	132	109	218

$$BP = \alpha_1 DBP + \alpha_2 SBP$$



PCA

Principal component analysis (PCA) is a method to find the linear combination that accounts for as much variability as possible.

$$\boxed{BP} = \alpha_1 DBP + \alpha_2 SBP$$

PCA

$$BP = \alpha_1 DBP + \alpha_2 SBP$$

Constraint:

$$\boxed{\alpha_1^2 + \alpha_2^2 + \ldots + \alpha_p^2 = 1}$$

$$\boxed{\alpha_1^2 + \alpha_2^2 = 1}$$

$$\boxed{0.8^2 + 0.6^2 = 0.64 + 0.36 = 1}$$

PCA

$$BP = 0.8DBP + 0.6SBP$$

Diastolic BP	Systolic BP	BP
78	126	138.0
80	128	140.8
81	127	141.0
82	130	143.6
84	130	145.2
86	132	148.0
	Mean =	**142.8**

$$\boxed{\mathrm{var}(Y) = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \overline{Y})^2} = \frac{1}{5}\left((138-142.8)^2 + \ldots + (148-142.8)^2\right) = 12.74$$

PCA

α_1	α_2	var(Y)
0.8	0.6	12.74
0.6	0.8	11.8
0.98	0.2	10.4
0.2	0.98	7.4

$$BP = 0.8DBP + 0.6SBP$$
$$BP = 0.6DBP + 0.8SBP$$
$$BP = 0.98DBP + 0.2SBP$$
$$BP = 0.2DBP + 0.98SBP$$

PCA

Diastolic BP	Systolic BP
78	126
80	128
81	127
82	130
84	130
86	132

	DBP	SBP
DBP	8.17	5.97
SBP	5.97	4.97

$$Eig = \begin{bmatrix} -0.8 \\ -0.6 \end{bmatrix}$$

$$BP = \boxed{-0.8}\,DBP + (\boxed{-0.6}\,SBP)$$

PCA

Diastolic BP	Systolic BP	Weight	Height
78	126	67	170
80	128	77	177
81	127	89	183
82	130	90	187
84	130	50	165
86	132	55	164

BP

BS

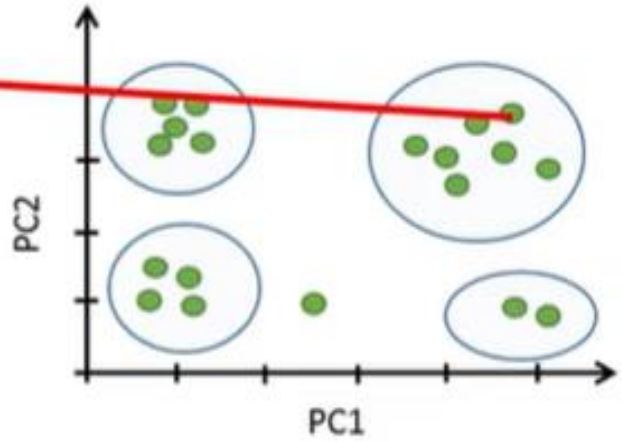
Cholesterol = BP + BS

PCA

Person	DBP	SBP	BMI	Chol.	Pulse	Temp	...
1	78	126	25	170	55	37.4	...
2	80	128	27	177	56	37.8	...
3	81	127	23	183	60	36.8	...
4	82	130	30	187	61	36.4	...
5	84	130	28	165	62	36.9	...
6	86	132	35	164	70	37.0	...
...	..	..	..	..	..	..	...

PCA

Person	PC1	PC2
1	3.4	4.5
2	3.7	4.4
3	-21.2	-15.2
4	-20.2	-16.5
5	-8.4	-8.9
6	-0.2	10.2
...	..	..

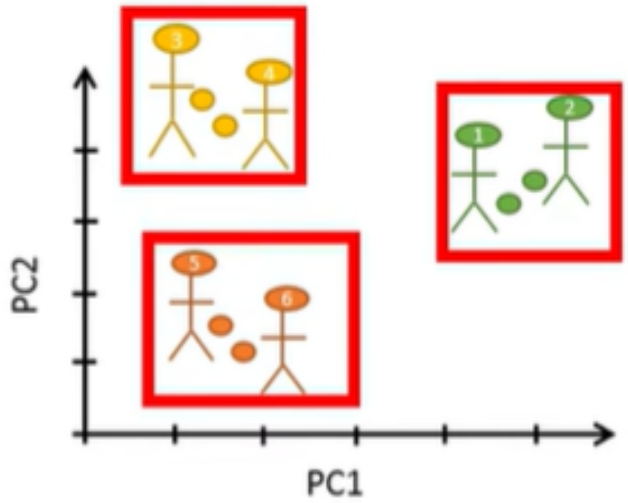


PCA

Person	Gene 1	Gene 2	Gene 3	Gene 4	.....
1	69	1	0	233	
2	70	2	0	231	
3	80	1	0	310	
4	70	0	0	288	
5	50	1	0	277	
6	60	2	0	235	

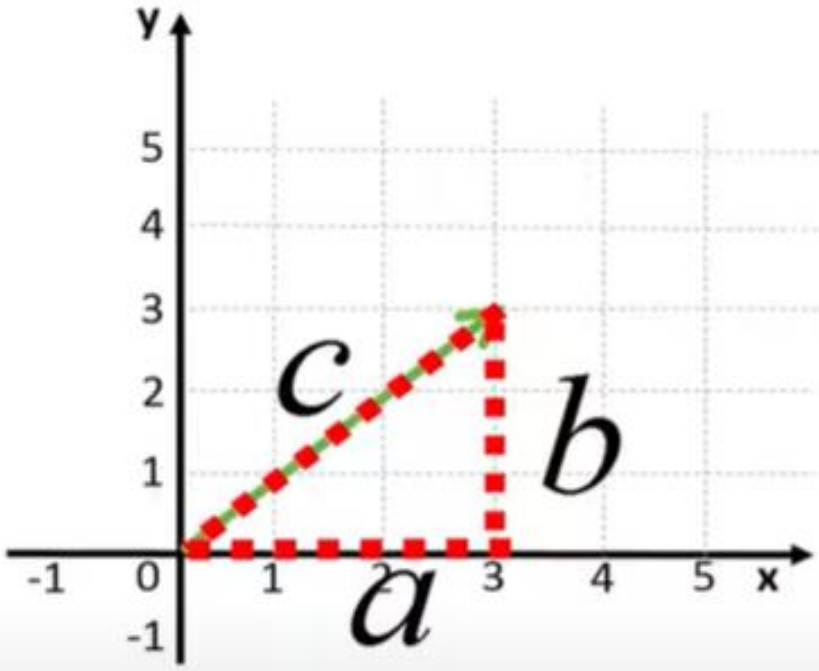
PCA

Person	PC1	PC2
1	8	2
2	9	3
3	-5	11
4	-4	10
5	-5	-12
6	-3	-14



Vectors

$$v = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

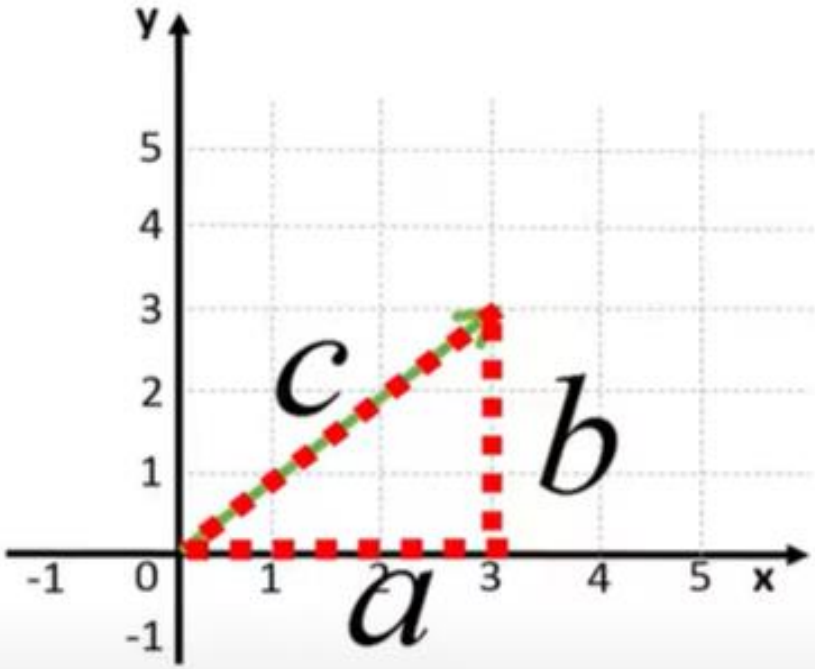


Pythagoras' theorem

$$c^2 = a^2 + b^2$$

Vectors

$$v = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$



$$c = \sqrt{a^2 + b^2} = \sqrt{3^2 + 3^2} = \sqrt{18}$$

Eigenvectors

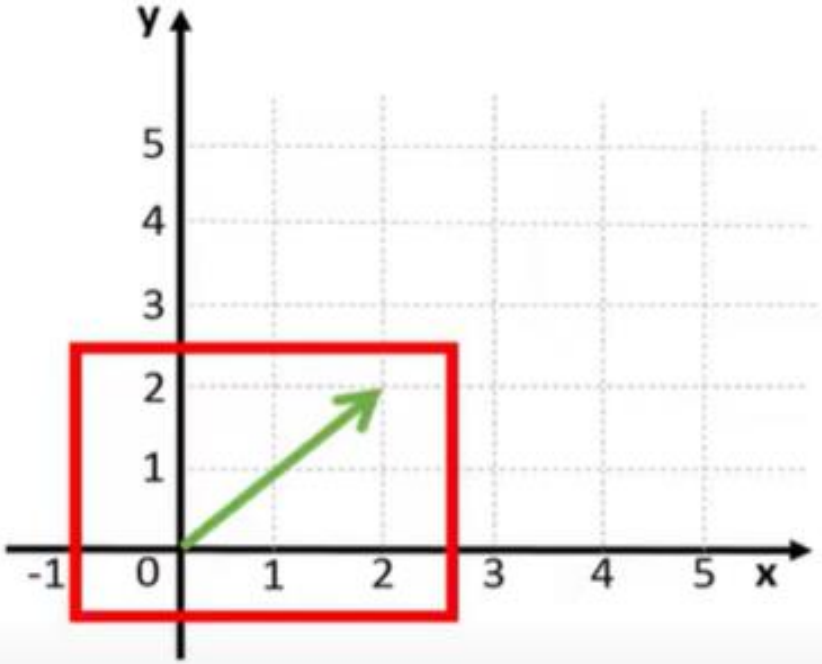
Eigenvectors

$$Av = \lambda v$$

Eigenvectors

$$v = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

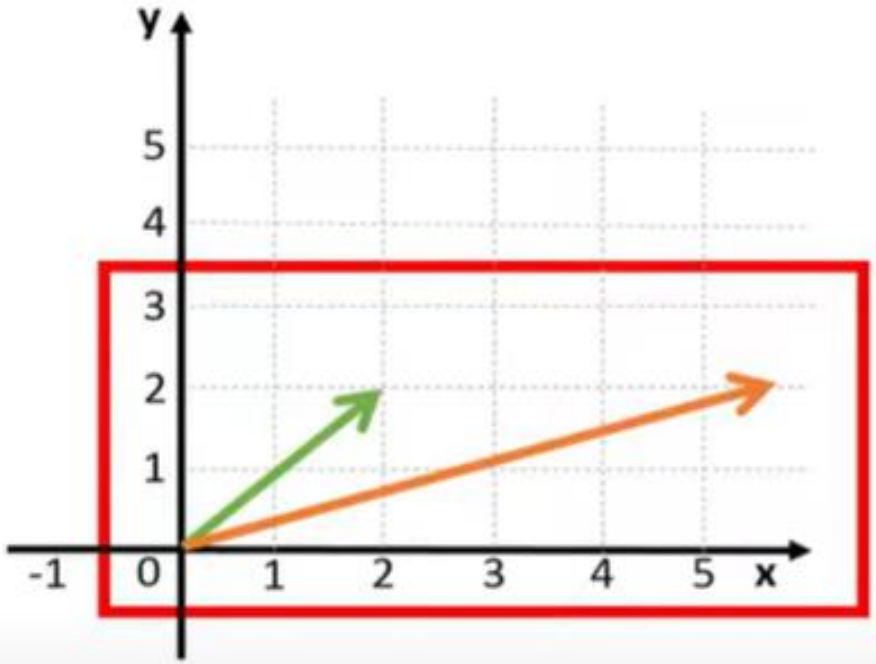
$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$



Eigenvectors

$$v = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$

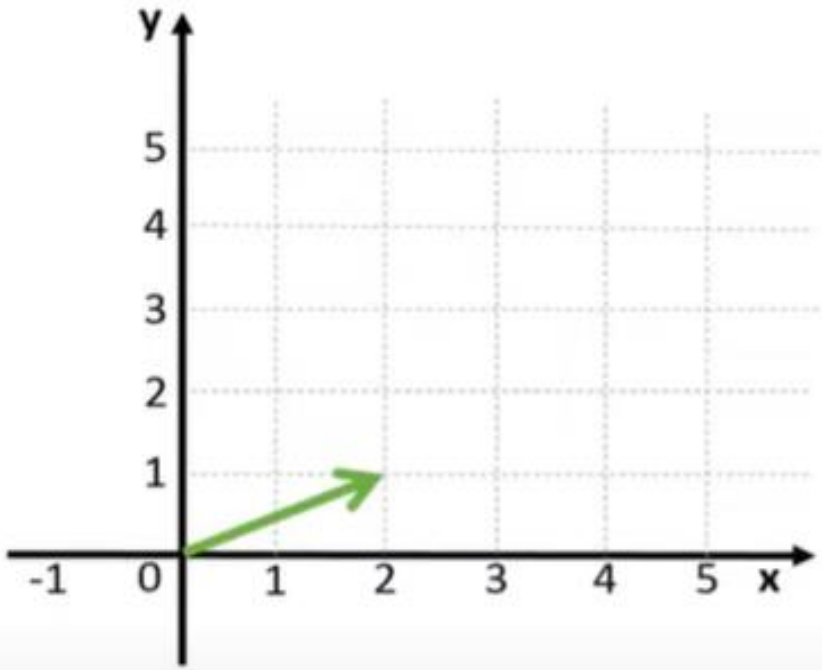


$$Av = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

Eigenvectors

$$v = \begin{bmatrix} 2 \\ \boxed{1} \end{bmatrix}$$

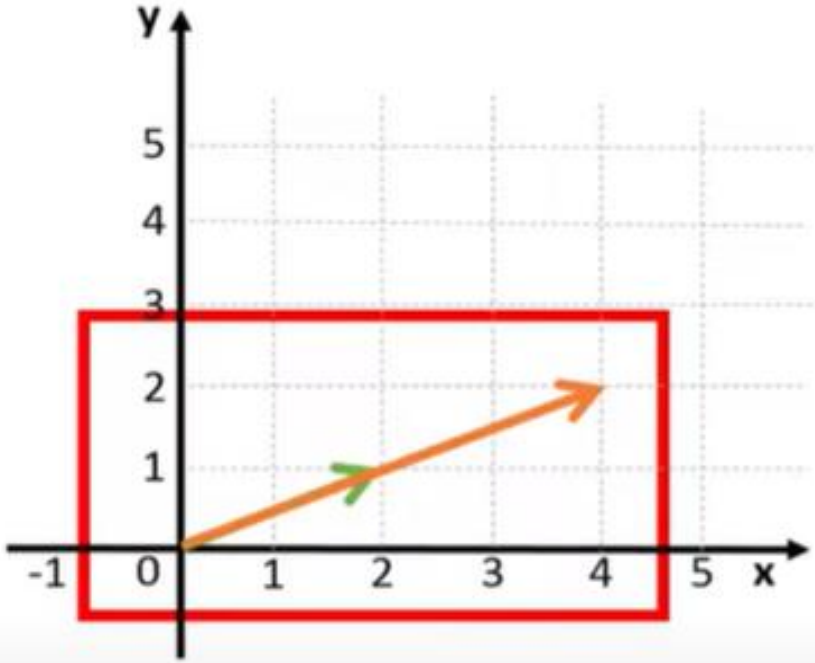
$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$



Eigenvalue

$$v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$



$$Av = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

Eigenvalue

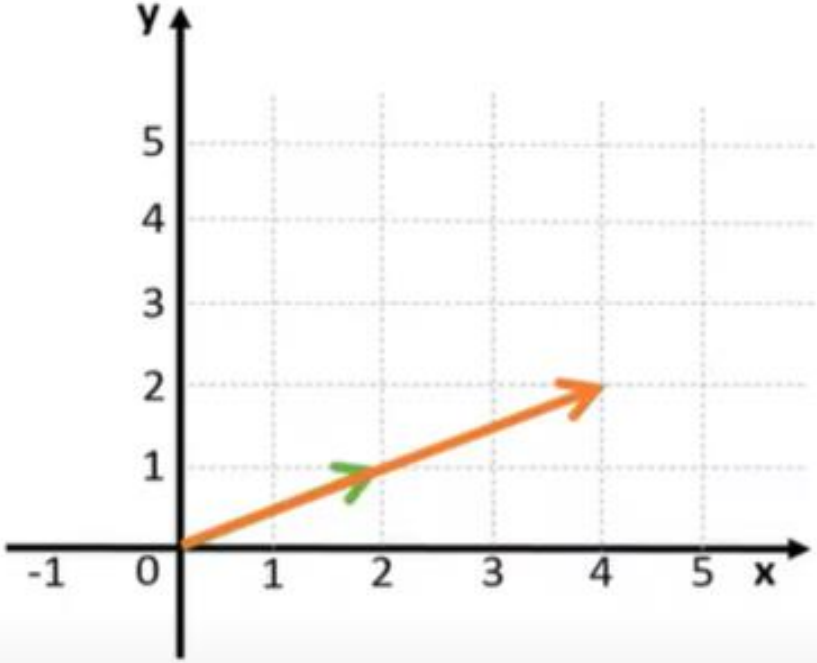
Eigenvalue

The eigenvalue tells us how much the eigenvector changes in size when multiplied with the matrix.

Eigenvalue

$$v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

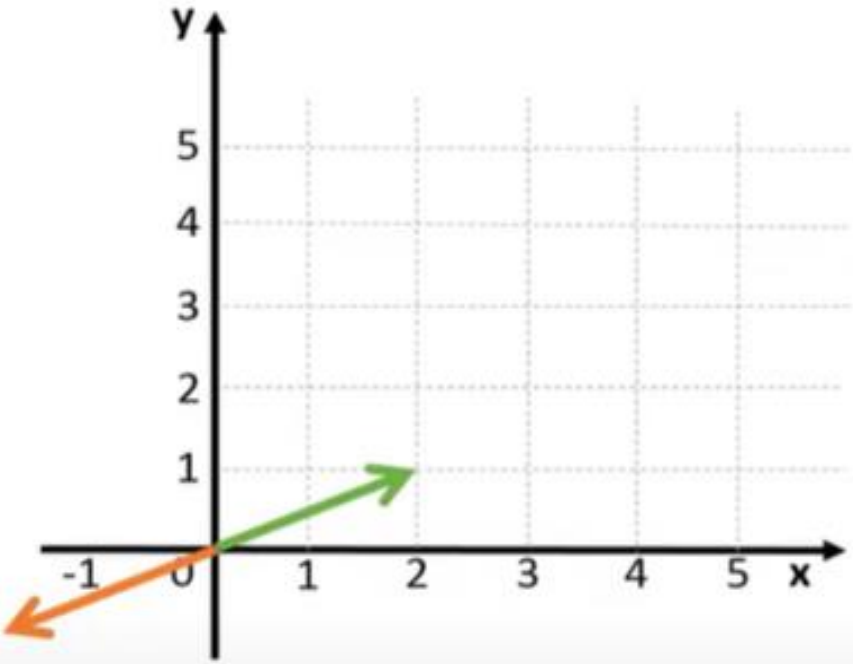
$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$



$$\lambda = 2$$

$$\boxed{Av} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 2\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \lambda v$$

Eigenvalue

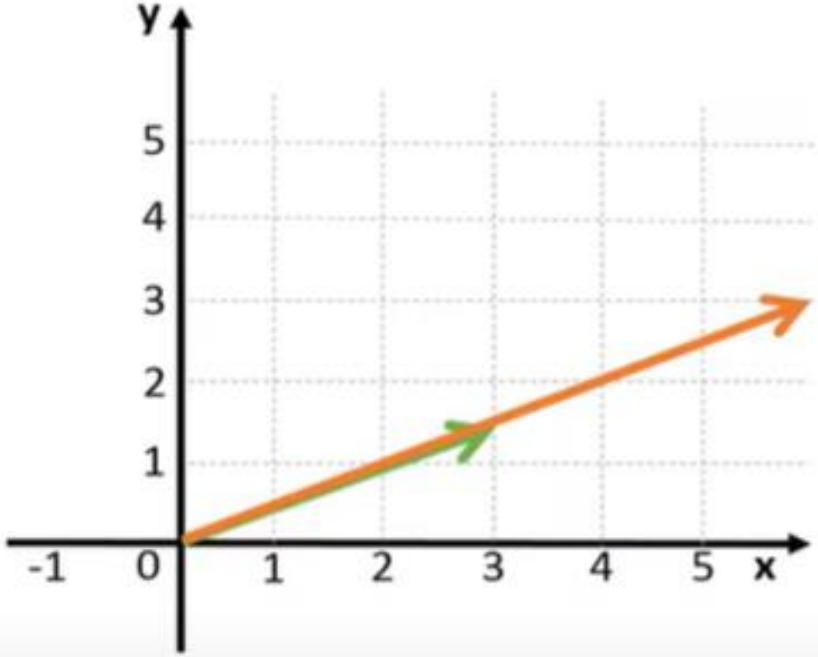


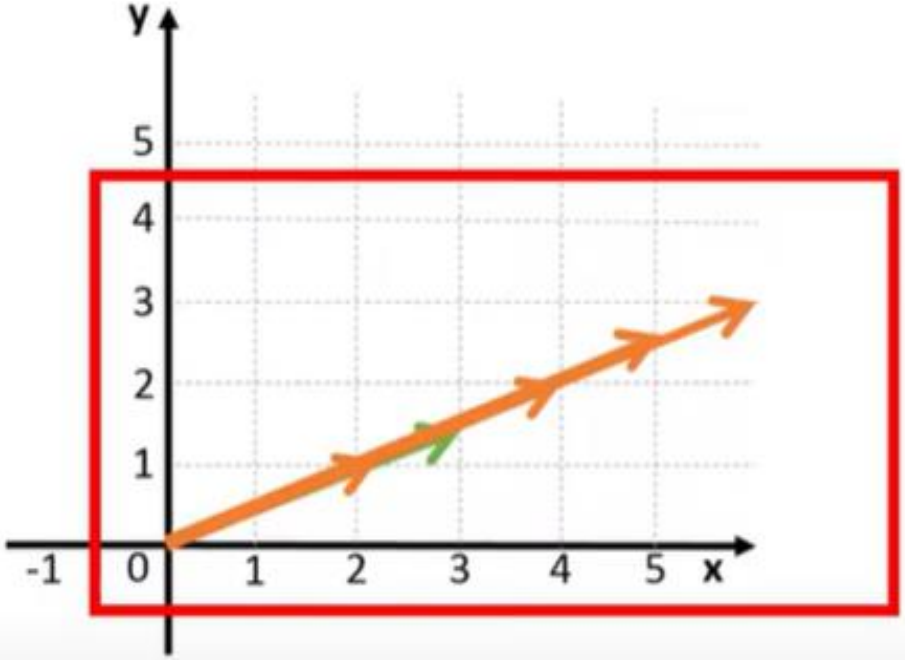
$$Av = \boxed{\lambda} v$$

Eigenvectors

$$v = \begin{bmatrix} 3 \\ 1.5 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$





$$Av = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 1.5 \end{bmatrix} = \boxed{\begin{bmatrix} 6 \\ 3 \end{bmatrix}} = 2\begin{bmatrix} 3 \\ 1.5 \end{bmatrix}$$

Unit length

$$v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\sqrt{2^2 + 1^2} = \sqrt{5}$$

$$2^2 + 1^2 = 5$$

$$\frac{2^2}{5} + \frac{1^2}{5} = \frac{5}{5}$$

$$\left(\frac{2}{\sqrt{5}}\right)^2 + \left(\frac{1}{\sqrt{5}}\right)^2 = \boxed{1}$$

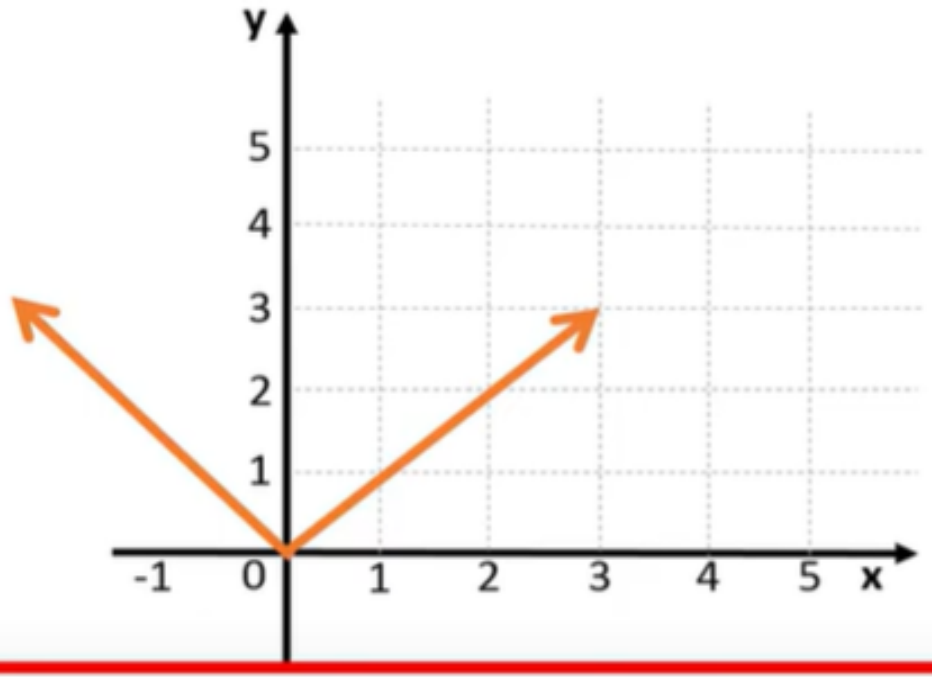
$$v = \boxed{\begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}} \approx \begin{bmatrix} 0.89 \\ 0.45 \end{bmatrix}$$

Eigenvectors

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

$$v_1 = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

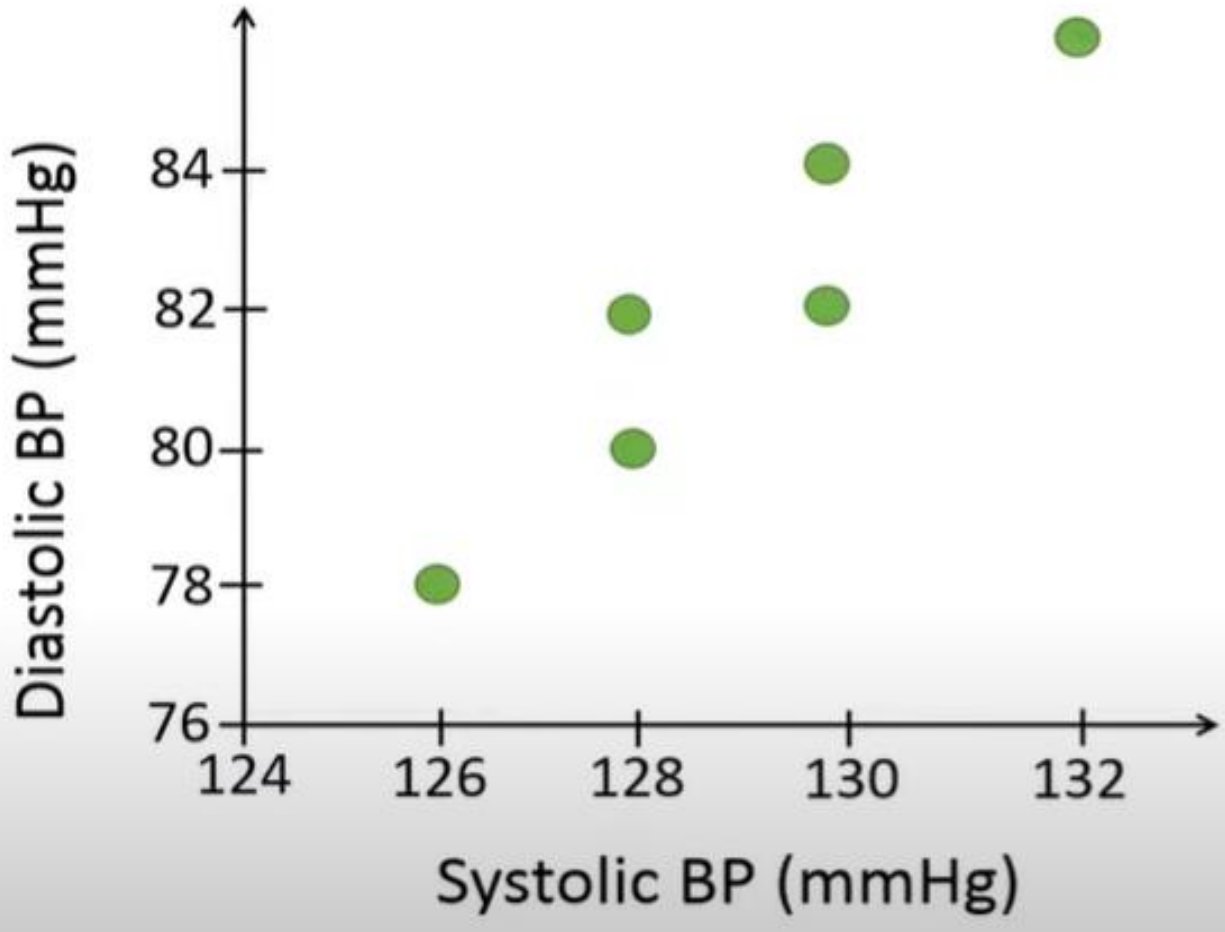
$$v_2 = \begin{bmatrix} -3 \\ 3 \end{bmatrix}$$



$$\boxed{v_1 v_2 = \begin{bmatrix} 3 & 3 \end{bmatrix} \cdot \begin{bmatrix} -3 \\ 3 \end{bmatrix} = -9 + 9 = 0}$$

Math behind PCA

Example data



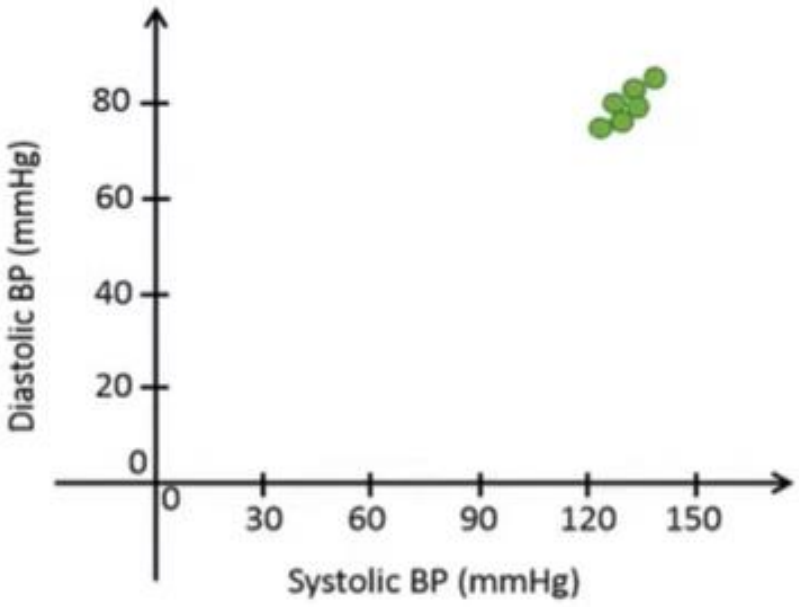
Systolic BP	Diastolic BP
126	78
128	80
128	82
130	82
130	84
132	86

PCA

1. Center the data
2. Calculate the covariance matrix
3. Calculate eigenvalues of the covariance matrix
4. Calculate eigenvectors of the covariance matrix
5. Order the eigenvectors
6. Calculate the principal components

1. Center the data

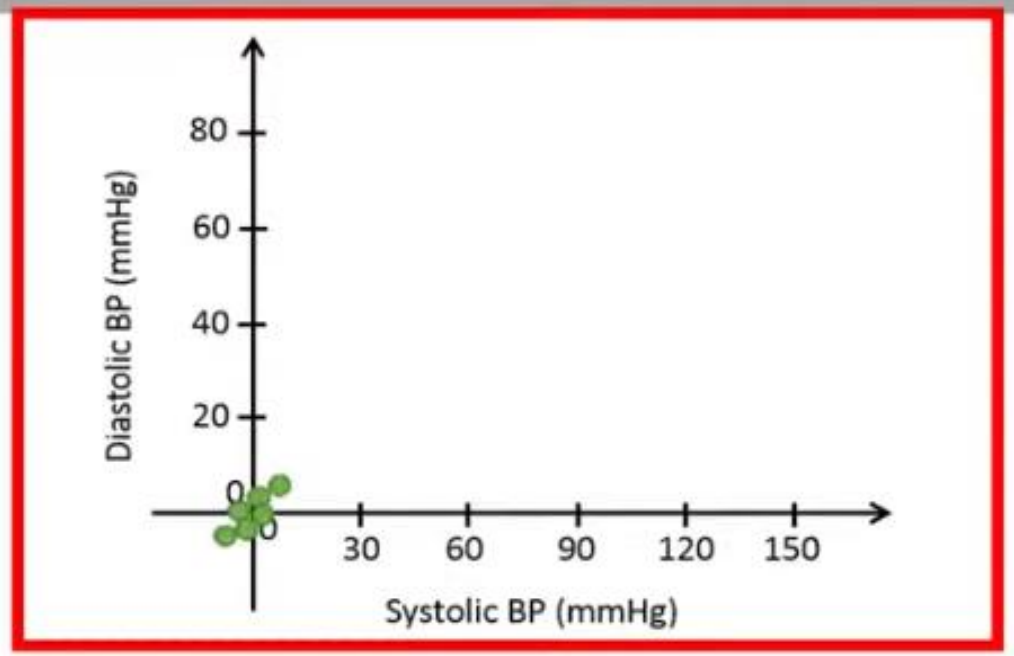
Systolic BP	Diastolic BP
126	78
128	80
128	82
130	82
130	84
132	86



1. Center the data

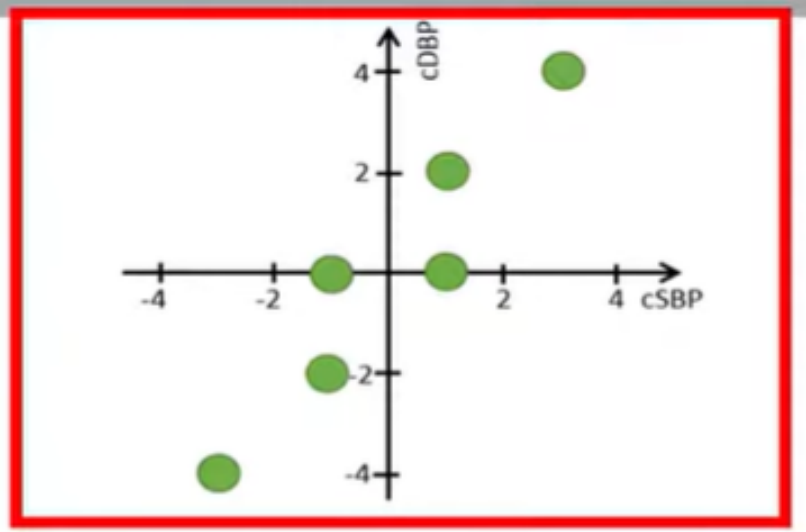
Systolic BP	Diastolic BP
126 -129 = -3	78 -82 = -4
128 -129 = -1	80 -82 = -2
128 -129 = -1	82 -82 = 0
130 -129 = 1	82 -82 = 0
130 -129 = 1	84 -82 = 2
132 -129 = 3	86 -82 = 4

Centered SBP	Centered DBP
-3	-4
-1	-2
-1	0
1	0
1	2
3	4



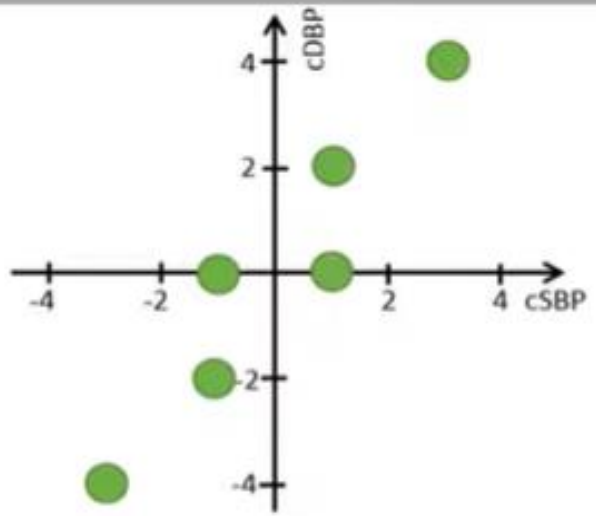
1. Center the data

Centered SBP	Centered DBP
-3	-4
-1	-2
-1	0
1	0
1	2
3	4



2. Calculate the covariance matrix

Centered SBP	Centered DBP
-3	-4
-1	-2
-1	0
1	0
1	2
3	4



	SBP	DBP
SBP	4.4	5.6
DBP	5.6	8.0

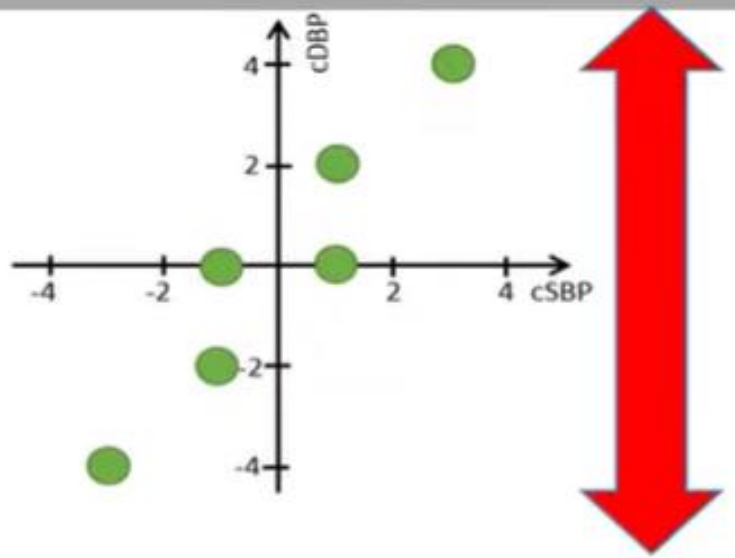
$$\mathrm{var}(\mathrm{cSBP}) = \frac{1}{n-1}\sum_{i=1}^{n}(\mathrm{cSBP}_i - \overline{\mathrm{cSBP}})^2 = ((-3)^2 + (-1)^2 + (-1)^2 + 1^2 + 1^2 + 3^2)/(6-1) = 22/5 = 4.4$$

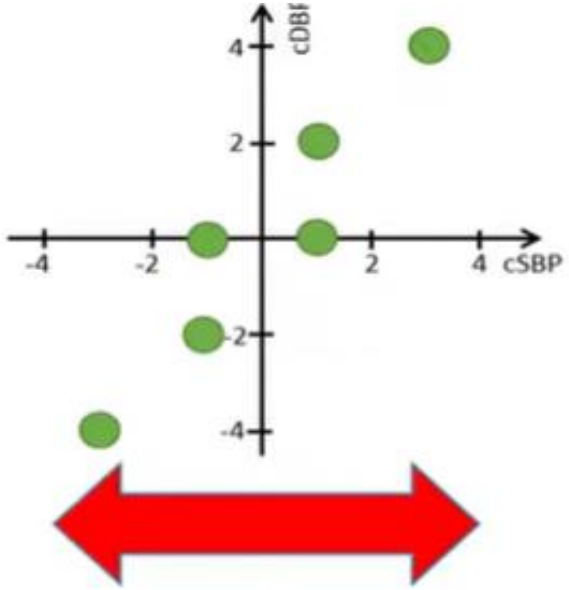
$$\mathrm{var}(\mathrm{cDBP}) = \frac{1}{n-1}\sum_{i=1}^{n}(\mathrm{cDBP}_i - \overline{\mathrm{cDBP}})^2 = ((-4)^2 + (-2)^2 + 0^2 + 0^2 + 2^2 + 4^2)/(6-1) = 40/5 = 8$$

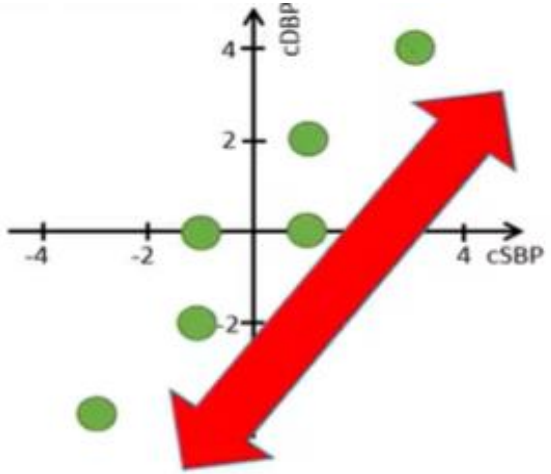
$$\boxed{\mathrm{cov}(\mathrm{cSBP},\mathrm{cDBP}) = \frac{1}{n-1}\sum(\mathrm{cSBP}_i - \overline{\mathrm{cSBP}})\cdot(\mathrm{cDBP}_i - \overline{\mathrm{cDBP}}) = ((-3)\cdot(-4) + (-1)\cdot(-2) + (-1)\cdot 0 + 1\cdot 0 + 1\cdot 2 + 3\cdot 4)/(6-1) = 28/5 = 5.6}$$

2. Calculate the covariance matrix

Centered SBP	Centered DBP
-3	-4
-1	-2
-1	0
1	0
1	2
3	4



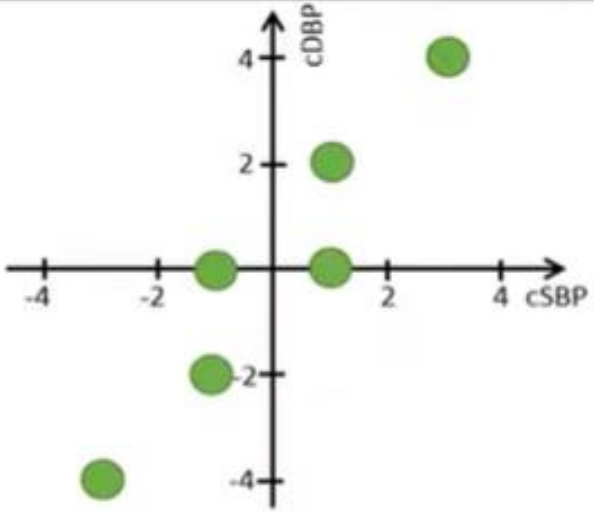




	SBP	DBP
SBP	4.4	5.6
DBP	5.6	8.0

3. Calculate the eigenvalues of the covariance matrix

Centered SBP	Centered DBP
-3	-4
-1	-2
-1	0
1	0
1	2
3	4



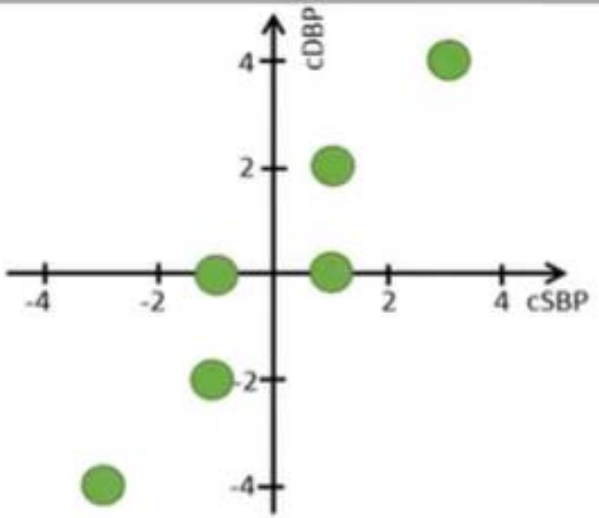
$$\det\left|\boxed{A} - \lambda I\right| = 0$$

	SBP	DBP
SBP	4.4	5.6
DBP	5.6	8.0

$$\det\left|\boxed{\begin{bmatrix} 4.4 & 5.6 \\ 5.6 & 8.0 \end{bmatrix}} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right| = 0$$

3. Calculate the eigenvalues of the covariance matrix

Centered SBP	Centered DBP
-3	-4
-1	-2
-1	0
1	0
1	2
3	4



	SBP	DBP
SBP	4.4	5.6
DBP	5.6	8.0

$$\det\left|A-\lambda I\right|=0$$

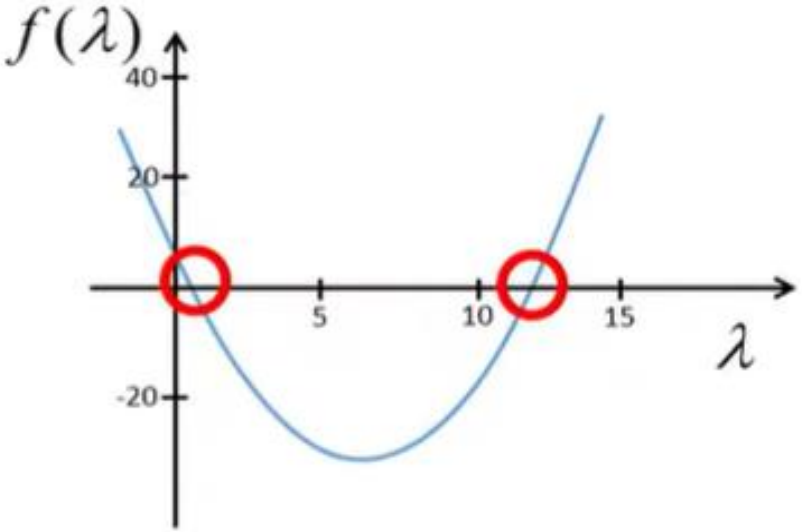
$$\det\left|\begin{bmatrix}(4.4-\lambda) & 5.6\\ 5.6 & (8.0-\lambda)\end{bmatrix}\right|=0$$

$$(4.4-\lambda)(8.0-\lambda)-5.6\cdot 5.6=0$$

$$\boxed{3.84-12.4\lambda+\lambda^2=0}$$

3. Calculate the eigenvalues of the covariance matrix

Centered SBP	Centered DBP
-3	-4
-1	-2
-1	0
1	0
1	2
3	4



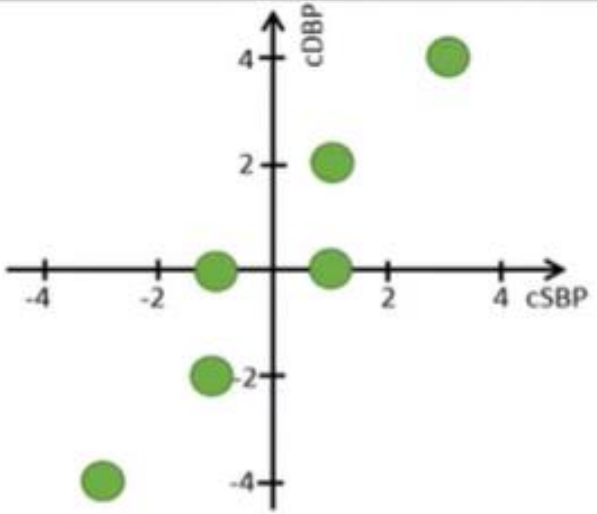
	SBP	DBP
SBP	4.4	5.6
DBP	5.6	8.0

$$\boxed{3.84 - 12.4\lambda + \lambda^2} = 0$$

$$\lambda_1 = 0.32 \qquad \lambda_2 = 12.08$$

4. Calculate the eigenvectors of the covariance matrix

Centered SBP	Centered DBP
-3	-4
-1	-2
-1	0
1	0
1	2
3	4



$$A \cdot v = \lambda \cdot v$$

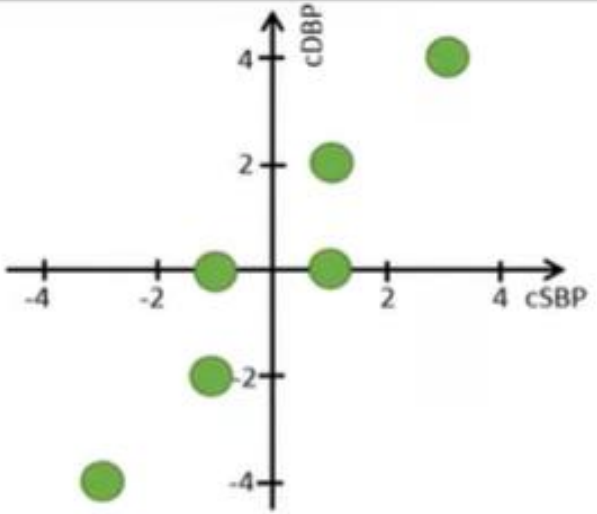
	SBP	DBP
SBP	4.4	5.6
DBP	5.6	8.0

$\lambda_1 = 0.32$

$\boxed{\lambda_2 = 12.08}$

4. Calculate the eigenvectors of the covariance matrix

Centered SBP	Centered DBP
-3	-4
-1	-2
-1	0
1	0
1	2
3	4



$$A \cdot v = \lambda \cdot v$$

	SBP	DBP
SBP	4.4	5.6
DBP	5.6	8.0

$$\begin{bmatrix} 4.4 & 5.6 \\ 5.6 & 8.0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = 12.08 \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\boxed{4.4x} + 5.6y = 12.08x$$

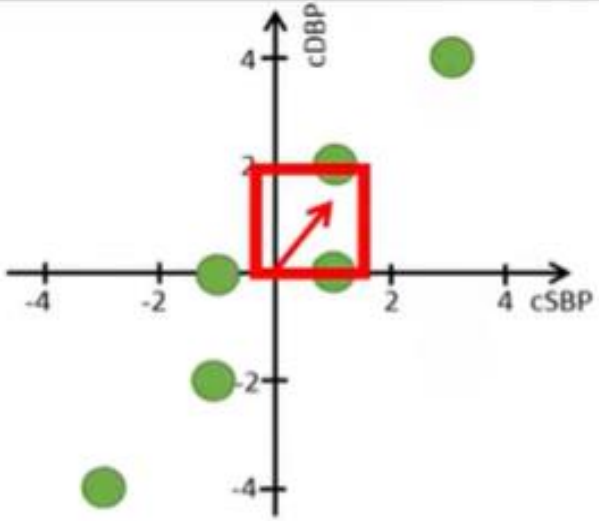
$$5.6x + \boxed{8.0y} = 12.08y$$

$$\boxed{\begin{aligned} 5.6y &= 7.68x \\ 5.6x &= 4.08y \end{aligned}}$$

$$\boxed{\begin{aligned} y &= 1.37x \\ 1.37x &= y \end{aligned}}$$

4. Calculate the eigenvectors of the covariance matrix

Centered SBP	Centered DBP
-3	-4
-1	-2
-1	0
1	0
1	2
3	4



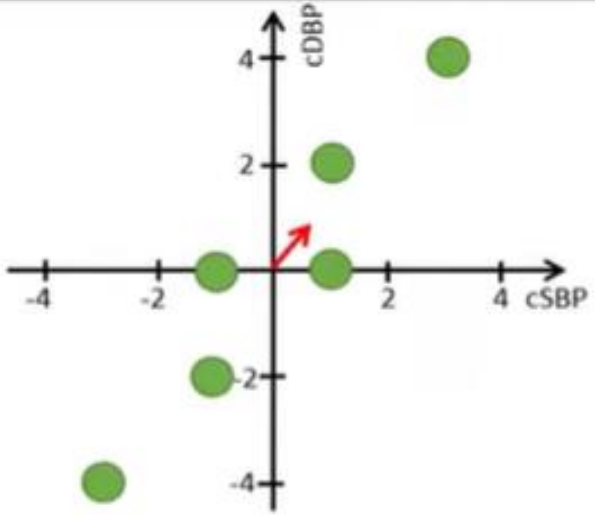
$$v_2 = \begin{bmatrix} 1 \\ 1.37 \end{bmatrix}$$

After Normalization $\boxed{v_2 = \begin{bmatrix} 0.59 \\ 0.81 \end{bmatrix}}$ $\lambda_2 = 12.08$

	SBP	DBP
SBP	4.4	5.6
DBP	5.6	8.0

4. Calculate the eigenvectors of the covariance matrix

Centered SBP	Centered DBP
-3	-4
-1	-2
-1	0
1	0
1	2
3	4



$$v_2 = \begin{bmatrix} 0.59 \\ 0.81 \end{bmatrix} \quad \lambda_2 = 12.08$$

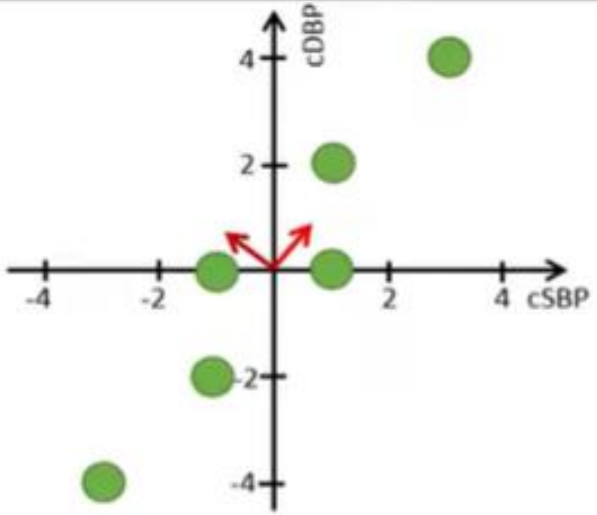
$$\lambda_1 = 0.32$$

	SBP	DBP
SBP	4.4	5.6
DBP	5.6	8.0

$$\begin{bmatrix} 4.4 & 5.6 \\ 5.6 & 8.0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \boxed{0.32} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

4. Calculate the eigenvectors of the covariance matrix

Centered SBP	Centered DBP
-3	-4
-1	-2
-1	0
1	0
1	2
3	4



$$v_2 = \begin{bmatrix} 0.59 \\ 0.81 \end{bmatrix} \quad \lambda_2 = 12.08$$

$$\boxed{v_1 = \begin{bmatrix} -0.81 \\ 0.59 \end{bmatrix}} \lambda_1 = 0.32$$

Order based on eigen values

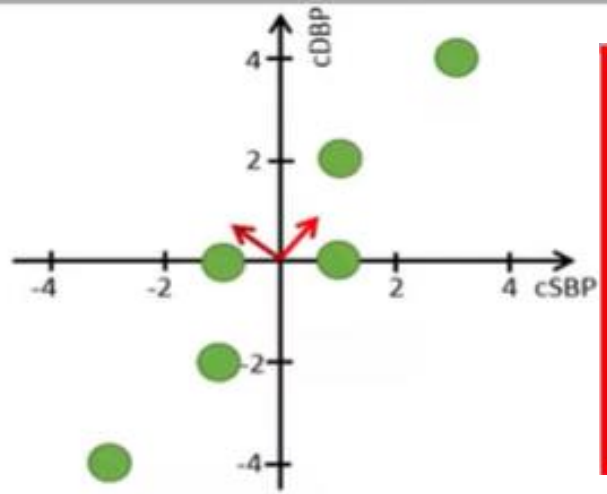
	SBP	DBP
SBP	4.4	5.6
DBP	5.6	8.0

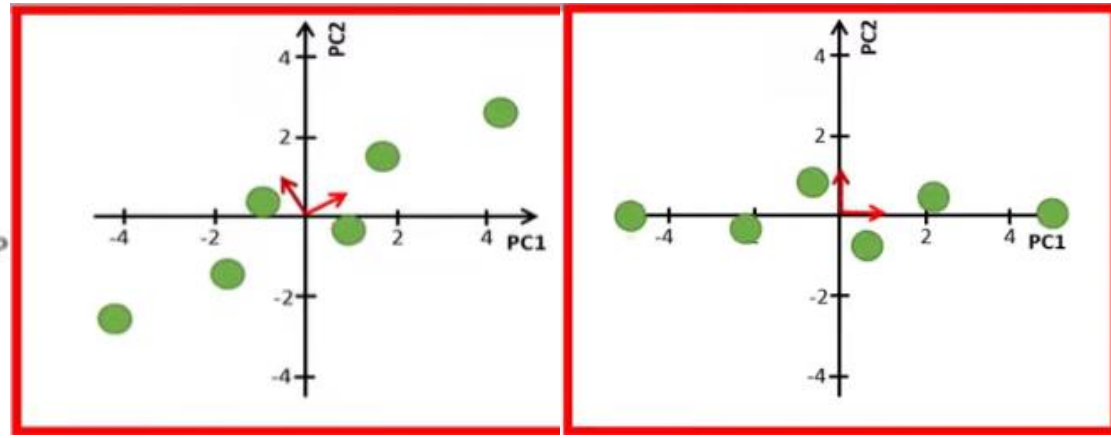
$$\boxed{v_1 = \begin{bmatrix} 0.59 \\ 0.81 \end{bmatrix} \quad \lambda_1 = 12.08}$$

$$v_2 = \begin{bmatrix} -0.81 \\ 0.59 \end{bmatrix} \quad \lambda_2 = 0.32$$

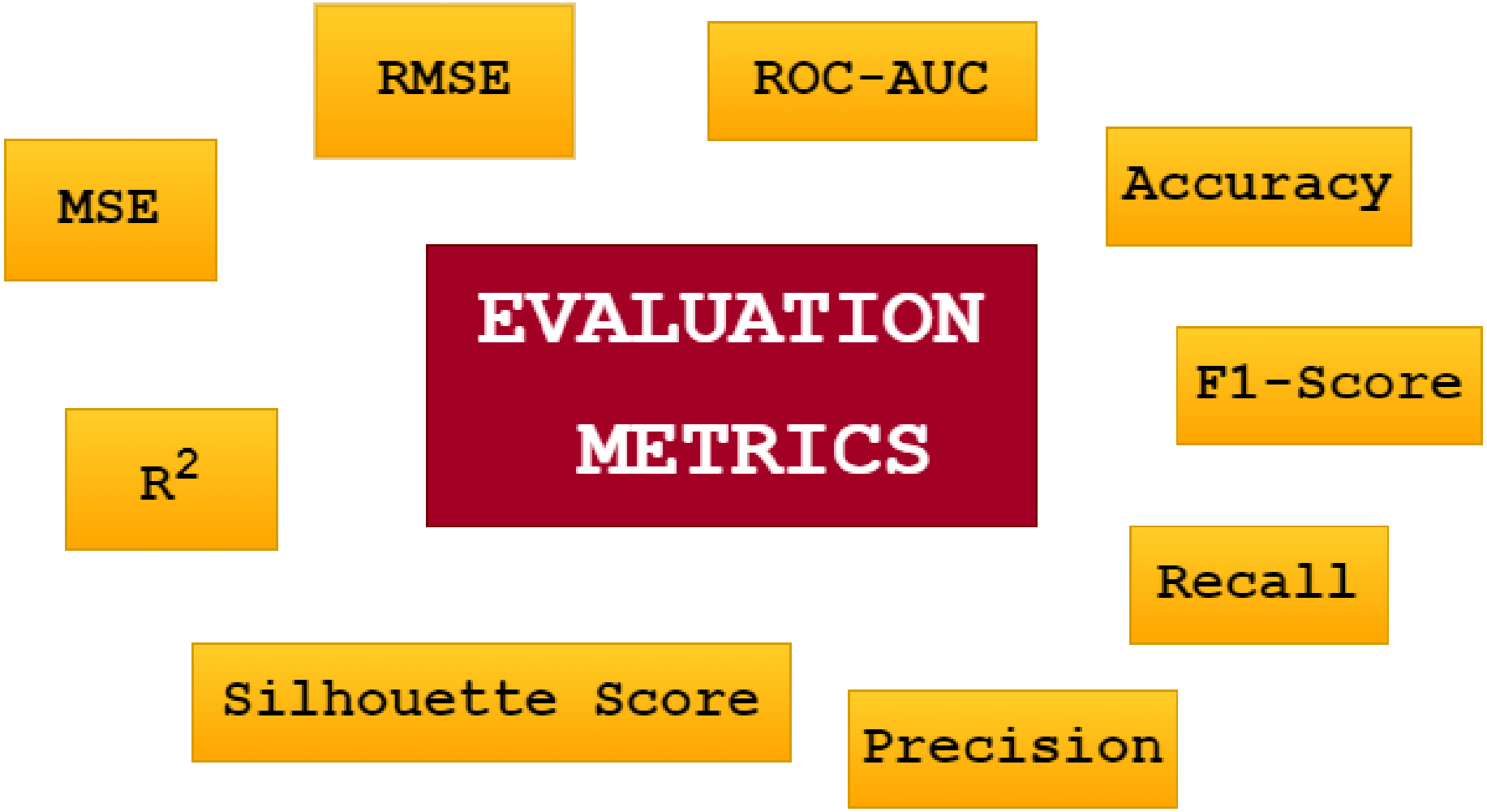
6. Calculate the principal components

Centered SBP	Centered DBP
-3	-4
-1	-2
-1	0
1	0
1	2
3	4





Principal Component Scores

$$V = \begin{bmatrix} 0.59 & -0.81 \\ 0.81 & 0.59 \end{bmatrix} \qquad D = \begin{bmatrix} -3 & -4 \\ -1 & -2 \\ -1 & 0 \\ 1 & 0 \\ 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad DV = \begin{bmatrix} -3 & -4 \\ -1 & -2 \\ -1 & 0 \\ 1 & 0 \\ 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 0.59 & -0.81 \\ 0.81 & 0.59 \end{bmatrix} = \boxed{\begin{bmatrix} -5.0 & 0.1 \\ -2.2 & -0.4 \\ -0.6 & 0.8 \\ 0.6 & -0.8 \\ 2.2 & 0.4 \\ 5.0 & -0.1 \end{bmatrix}}$$

Evaluation Metrics in Machine Learning